PROOF OF THE DURFEE CONJECTURE FOR TWO-DIMENSIONAL HYPERSURFACE SINGULARITIES

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Abstract. We prove the Durfee conjecture for two-dimensional isolated hypersurface singularities: the Milnor number of such a singularity is strictly larger than six times the geometric genus.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Let $f \in \mathbb{C}\{x, y, z\}$ with $f \neq 0$ and $f(0) = 0$, and assume that the surface germ $(X, 0) = (f = 0) \subset (\mathbb{C}^3, 0)$ is smooth or has an isolated singularity at 0. The Milnor number of f is

$$\mu(f, 0) = \dim_{\mathbb{C}} \mathbb{C}\{x, y, z\}/(\partial f/\partial x, \partial f/\partial y, \partial f/\partial z).$$

We let $\pi : Y \to X$ be the minimal resolution of a small representative of $(X, 0)$. The geometric genus of $(X, 0)$ is

$$p_g(X, 0) = \dim_{\mathbb{C}}(R^1\pi_*\mathcal{O}_Y)_0, \tag{1.1}$$

which does not depend on the choice of the resolution.

Durfee [Dur78] considered 2-dimensional complete intersection germs with at most an isolated singularity. Such a germ $(X, 0)$ is normal, so its geometric genus $p_g = p_g(X, 0)$ is defined by (1.1), and the Milnor number μ of such a germ is the second Betti number of the Milnor fiber [Dur78, §2]. For the hypersurface germ $(X, 0) = (f = 0)$, $\mu = \mu(f, 0)$ [Dur78, §2, (1)]. Durfee proposed the following conjecture.

Conjecture 1.1 ([Dur78, Conjecture 5.3])**.** *Let $(X, 0)$ be a 2-dimensional complete intersection germ with at most an isolated singularity. Then the Milnor number μ and the geometric genus p_g of $(X, 0)$ satisfy $6p_g \leq \mu$, and $6p_g = \mu$ only if $\mu = 0$.*

By [Dur78, p. 97], a germ $(X, 0)$ as in Conjecture 1.1 is smooth if and only if $\mu = 0$. Thus, for singular $(X, 0)$, Conjecture 1.1 is the strict inequality $6p_g < \mu$.

Date: November 10, 2026.

2020 *Mathematics Subject Classification.* 32S25, 14J17, 14B05, 13H15, 13P10.

Key words and phrases. Durfee conjecture, Milnor number, geometric genus, hypersurface singularity, surface singularity, colength of products of ideals, initial ideal.

By [KN12, §3] and [KN13, §1.2, II], there exist 2-dimensional complete intersection germs of codimension ≥ 2 with an isolated singularity on which $6p_g \leq \mu$ fails. Kerner and Némethi [KN13, §1.1] and Kollár and Némethi [KN17b, p. 788] call $6p_g \leq \mu$ the strong inequality, and Enokizono calls the inequality $6p_g \leq \mu$ for hypersurface germs Durfee's strong conjecture [Eno17, Introduction].

Let $\nu = \max\{j \mid f \in (x,y,z)^j\}$ be the multiplicity of f, let $E = \pi^{-1}(0)_{\mathrm{red}}$, let n be the number of π-exceptional curves, and let $b_1(E) = \dim_{\mathbb{C}} H^1(E,\mathbb{C})$. The inequality $6p_g(X,0) \leq \mu(f,0)$ was known in the following cases:

(1) When $\nu = 2$ [Tom93].
(2) When $\nu = 3$ [Ash92].
(3) When f is quasi-homogeneous [XY93].
(4) When $f = g(x,y) + z^k$ for some $g \in \mathbb{C}\{x,y\}$ and some positive integer k, and either g is irreducible [Nem98] or k is coprime to the multiplicity of the pullback of g along any exceptional curve of the minimal embedded resolution of $(g=0) \subset (\mathbb{C}^2,0)$ [Nem99].
(5) When $(X,0)$ is absolutely isolated, that is, blowing up the singular points finitely many times resolves $(X,0)$ [MH00].
(6) When the link of $(X,0)$ is an integral homology sphere [KN17b].
(7) When f is Newton non-degenerate with respect to a convenient Newton diagram $\ell\Gamma$, where Γ is a fixed Newton diagram and ℓ is a sufficiently large positive integer [KN17a].
(8) When $b_1(E) \leq n+1$ [Eno17].

Kollár and Némethi [KN17b, Theorem 4(2)] proved an upper bound of the signature of the Milnor fiber of $(X,0)$. By [KN17b, (1)] and [Dur78, Proposition 3.1 and the paragraph before it], together with the invariance of $b_1(E)$ under point blow-ups (see the proof of Lemma 2.3), their bound is equivalent to

$$\mu(f,0) + b_1(E) \geq 5p_g(X,0) + n.$$

They wrote that the inequality $6p_g(X,0) \leq \mu(f,0)$ was still open in general [KN17b, p. 789], and Kerner and Némethi could not prove it for an arbitrary Newton diagram, even when f is Newton non-degenerate [KN17a, §1.4].

By [EF26, §2.2], if $(X,0)$ is singular, then the surface case of the conjecture [EF26, Conjecture B] of Eriksson and Freixas i Montplet is the strict inequality $6p_g(X,0) < \mu(f,0)$, and the cases of [EF26, Conjecture B] for isolated hypersurface singularities of dimension at least 3 are open [EF26, §1.3].

The goal of this paper is to prove Conjecture 1.1, including the equality case, when $(X,0)$ is a hypersurface germ.

Theorem 1.2. *Let $f \in \mathbb{C}\{x,y,z\}$ with $f \neq 0$ and $f(0) = 0$, and assume that the surface germ $(X,0) = (f=0) \subset (\mathbb{C}^3,0)$ is smooth or has an isolated singularity at 0. Then the Milnor number $\mu(f,0)$ and the geometric genus $p_g(X,0)$ satisfy*

$$6p_g(X,0) \leq \mu(f,0),$$

and $6p_g(X,0) = \mu(f,0)$ if and only if $(X,0)$ is smooth.

Sketch of the proof. Since $\mu(f,0) = p_g(X,0) = 0$ if $(X,0)$ is smooth and $\mu(f,0) \geq 1$ if $(X,0)$ is singular, it suffices to prove $\mu - 6p_g > 0$ when $(X,0)$ is singular and $p_g > 0$, where $\mu = \mu(f,0)$ and $p_g = p_g(X,0)$. We use the notation of Definition 2.1, Set-up 2.2 and Definition 3.3: $R = \mathcal{O}_{X,0}$ with maximal ideal $\mathfrak{m}$, λ is the colength, Z_K the anti-canonical cycle, $I(D) = (\pi_*\mathcal{O}_Y(-D))_0$, $I_K = I(Z_K)$ the adjoint ideal, M the maximal ideal cycle, and $c(Q)$, N_Q are as in Definition 3.3. By [KN17b, Lemma 7], there exists an exceptional cycle l such that

$$\frac{1}{2}Z_K \leq l \leq Z_K \quad \text{and} \quad \chi(l) = \min\{\chi(D) \mid D \text{ is an exceptional cycle}\},$$

where $\chi(D) = -D \cdot (D - Z_K)/2$. Let

$$I = I(l), \quad J = I(Z_K - l), \quad \kappa = 4(\lambda(I) + \lambda(J)) - \lambda(IJ) - 1.$$

Let $q = n + \max\{(Z_K - 2D)^2 \mid D \text{ is an exceptional cycle}\}$ (Definition 5.3). Since $(Z_K - 2D)^2 = Z_K^2 - 8\chi(D)$ for any exceptional cycle D, $q = n + (Z_K - 2l)^2$ (see (6.1)). Laufer's formula (Lemma 2.3), the inequality $\lambda(I) + \lambda(J) \le p_g + \chi(l)$ of Lemma 2.4 for $D = l$ (cf. [KN17b, (9) and (10)]), and $\lambda(IJ) \ge \lambda(I_K) = p_g$ imply

$$\mu - 6p_g \ge 2 + q - b_1(E) + 2\kappa$$

(see (6.2)). If $l \ne Z_K$, then I and J are $\mathfrak{m}$-primary, and $I \subset J$ as $l \ge Z_K - l$. Then

$$\begin{aligned} \kappa + 1 &\ge c(IJ) + N_I + \lceil \sqrt{\lambda(I) - \lambda(J)} \rceil && \text{(Proposition 3.4)}, \\ c(IJ) &\ge c(I_K) = h^1(\mathcal{O}_M) && \text{(Lemma 4.1)}, \\ q + 2h^1(\mathcal{O}_M) - b_1(E) &\ge 1 - \nu && \text{(Proposition 5.4(2))}. \end{aligned}$$

Hence, as $N_I \ge r = \lceil (\nu - 2)/2 \rceil$ by Proposition 4.2 (see (6.4)),

$$2 + q - b_1(E) + 2\kappa \ge 1 - \nu + 2N_I + 2\lceil \sqrt{\lambda(I) - \lambda(J)} \rceil > 0$$

unless $N_I = r$ and $\lambda(I) = \lambda(J)$ (see (6.3)).

If $l = Z_K$, then $J = R$, and Proposition 5.4(2), $h^1(\mathcal{O}_M) \le p_g$ (Lemma 4.1) and $p_g \ge \max\{1, \binom{\nu}{3}\}$ (Proposition 4.2) give $2 + q - b_1(E) + 2\kappa \ge 1 - \nu + 4p_g > 0$ (**Step 2** in Section 6). If $N_I = r$ and $\lambda(I) = \lambda(J)$, then $I = J = \mathfrak{m}^r$ (see (6.5)), so $\mathfrak{m}^{2r} = IJ \subset I_K$ and $Z_K \le 2rM$. When $\nu = 2r + 1$, Lemma 5.2 turns $Z_K \le 2rM$ into $-Z_K \cdot E \le 2r\nu$, and Proposition 5.4(1) gives $2 + q - b_1(E) + 2\kappa > 0$ (**Step 4**). When $\nu = 2r + 2$, Proposition 4.2 gives $I_K = \mathfrak{m}^{2r}$, and Lemma 5.2 turns $Z_K \le 2rM$ into $Z_K^2 \ge -4r^2\nu$, which with Lemma 2.3 and $b_1(E) \le 2h^1(\mathcal{O}_E) \le 2h^1(\mathcal{O}_M)$ (Lemma 5.1 and $M \ge E$) gives $\mu - 6p_g > 0$ (**Step 5**).

Proposition 3.4 is proved by passing to the initial monomial ideals $I' \subset J'$ of I, J in suitable linear coordinates, which keeps λ, and $\lambda(I' + (x, y)) = N_I$. For a monomial ideal Q of finite colength, let

$$\varphi_Q(i, j) = \min\{k \mid x^i y^j z^k \in Q\} \quad \text{for } (i, j) \in \mathbb{N}^2.$$

The inequality

$$\varphi_{I'J'}(i + i', j + j') \le \varphi_{I'}(i, j) + \varphi_{J'}(i', j') \quad \text{for } |i - i'| \le 1 \text{ and } |j - j'| \le 1$$

gives Lemma 3.1, whose last term $\lceil \sqrt{\lambda(I') - \lambda(J')} \rceil$ depends on

$$\sum_{p,p' \text{ adjacent}} |\delta(p) - \delta(p')| \ge 2\sqrt{\lambda(I') - \lambda(J')}, \quad \text{where } \delta = \varphi_{I'} - \varphi_{J'}.$$

Lemma 3.2 on $\mathbb{C}[x, y, z]/(x)$ and $\mathbb{C}[x, y, z]/(y)$ gives the terms $N_I + c(IJ)$.

Proposition 3.4 replaces de Fernex's inequality [KN17b, Lemma 26], which Kollár and Némethi use in the proof of [KN17b, Proposition 8] and which for R reads $4(\lambda(I) + \lambda(J)) > \lambda(IJ)$, that is, $\kappa \ge 0$. From $\kappa \ge 0$, (6.2), and Proposition 5.4(1), whose proof follows [KN17b, (16)–(19)], we obtain $\mu - 6p_g \ge 1 + Z_K \cdot E$, which is [KN17b, Proposition 20] for hypersurface germs, rewritten by Laufer's formula. Since $\lambda(I_K) = p_g > 0$, $0 \ne Z_K \ge 0$, so $Z_K^2 < 0$ and $Z_K \cdot E_i < 0$ for some i. As $Z_K \cdot E_i = -K_Y \cdot E_i \le 0$ for any i by the minimality of π, $1 + Z_K \cdot E \le 0$, so $\mu - 6p_g \ge 1 + Z_K \cdot E$ does not give $\mu > 6p_g$.

Structure of the paper. Section 2 fixes the notation and recalls Laufer's formula and the duality for complementary cycles. Section 3 bounds from above the colength of a product of two ideals, one contained in the other, in embedding dimension at most 3. Section 4 computes the colength of a general linear section of the adjoint ideal and bounds from below the order of the elements of the adjoint ideal. Section 5 proves two lower bounds for $q - b_1(E)$. Section 6 proves the main theorem.

Acknowledgements. The work was partially supported by the National Key R&D Program of China #2024YFA1014400. Jihao Liu would like to thank Bin Dong, Guoxiong Gao, Zeming Sun, Bin Wu, and Leheng Chen for AI assistance. Jihao Liu would like to thank Ruochuan Liu and Gang Tian for consistent support and encouragement.

2. Preliminaries

Definition 2.1. Let $S = \mathbb{C}\{x, y, z\}$ with maximal ideal $\mathfrak{n} = (x, y, z)$, let $\mathfrak{a} \subset \mathfrak{n}$ be an ideal, and let $R = S/\mathfrak{a}$ and $\mathfrak{m} = \mathfrak{n}/\mathfrak{a}$. For a $\mathbb{C}$-algebra A and an ideal $Q \subset A$ with $\dim_{\mathbb{C}} A/Q < +\infty$, let $\lambda(Q) = \dim_{\mathbb{C}} A/Q$.

Set-up 2.2. Let f, $(X, 0)$, $\mathfrak{a}$, ν, π, $E_1, \dots, E_n$, E, and $b_1(E)$ be as follows, with the notation of Definition 2.1.

(1) $f \in \mathfrak{n}^2$ is a power series such that the surface germ $(X, 0) = (f = 0) \subset (\mathbb{C}^3, 0)$ has an isolated singularity at 0, and $\mathfrak{a} = (f)$, so that $R = \mathcal{O}_{X,0}$.
(2) $\nu = \operatorname{mult}_0 f$, where $\operatorname{mult}_0 u = \max\{j \mid u \in \mathfrak{n}^j\}$ for $u \in S\backslash\{0\}$.
(3) $\pi : Y \to X$ is the minimal resolution of a small Stein representative X of the germ.
(4) $E_1, \dots, E_n$ are the irreducible components of $\pi^{-1}(0)$, $E = E_1 + \cdots + E_n$, and $b_1(E) = \dim_{\mathbb{C}} H^1(E, \mathbb{C})$.

An *exceptional cycle* is a divisor $D = \sum_i d_i E_i$ with $d_i \in \mathbb{Z}$. We write $D \geq D'$ if $d_i \geq d_i'$ for any i. Let Z_K be the exceptional cycle with $Z_K \cdot E_i = -K_Y \cdot E_i$ for any i. We call Z_K the *anti-canonical cycle*. For an exceptional cycle D, let $\chi(D) = -D \cdot (D - Z_K)/2$. For an exceptional cycle $D \geq 0$, let $I(D) = (\pi_* \mathcal{O}_Y(-D))_0 \subset R$. We call $I_K = I(Z_K)$ the *adjoint ideal* (cf. [MT80, Définition 1.6]). For $u \in R\backslash\{0\}$, let $v_i(u) = \operatorname{mult}_{E_i} \pi^* u$. The *maximal ideal cycle* is $M = \sum_i m_i E_i$, where $m_i = \min\{v_i(u) \mid u \in \mathfrak{m}\backslash\{0\}\}$.

Since π is minimal, $Z_K \cdot E_i = -K_Y \cdot E_i \leq 0$ for any i, so $Z_K \geq 0$ by the negativity lemma, and I_K is defined. By the Riemann–Roch formula, $\chi(\mathcal{O}_D) = \chi(D)$ for any exceptional cycle $D \geq 0$, both sides being 0 if $D = 0$.

Lemma 2.3 (cf. [Lau77] and [Dur78, Theorem 2.2])**.** *Let f, $(X, 0)$, π, $E_1, \dots, E_n$, E, $b_1(E)$, and Z_K be as in Set-up 2.2. Then*

$$\mu(f, 0) - 6p_g(X, 0) = 6p_g(X, 0) + Z_K^2 + n - b_1(E).$$

Proof. By [Dur78, Theorem 2.2], $\mu(f, 0) = 12p_g(X, 0) + K_{Y'}^2 + n' - b_1(E')$ for a good resolution $Y' \to X$ factoring through π, where n', $b_1(E')$, and the cycle Z_K' are defined as for π and $K_{Y'}^2 = Z_K'^2$. Since a point blow-up decreases K^2 by 1, increases n' by 1 and keeps $b_1(E')$, $\mu(f, 0) = 12p_g(X, 0) + Z_K^2 + n - b_1(E)$. The lemma follows. □

Lemma 2.4 (cf. [MT80, p. 239, footnote] and [KN17b, (9) and (10)])**.** *Notation and conditions as in Set-up 2.2. Then $\lambda(I_K) = p_g(X, 0)$, and*

$$\lambda(I(D)) + \lambda(I(Z_K - D)) \leq p_g(X, 0) + \chi(D)$$

for any exceptional cycle D with $0 \leq D \leq Z_K$.

Proof. Since $\omega_Y \cong \mathcal{O}_Y(-Z_K)$, Grauert–Riemenschneider vanishing implies

$$R^1 \pi_* \mathcal{O}_Y(-Z_K) = 0.$$

Since $\dim \pi^{-1}(x) \leq 1$ for any $x \in X$, $R^2 \pi_* \mathcal{O}_Y(-Z_K) = 0$. Taking stalks at 0 of the direct images of

$$0 \to \mathcal{O}_Y(-Z_K) \to \mathcal{O}_Y \to \mathcal{O}_{Z_K} \to 0,$$

where $\mathcal{O}_{Z_K} = 0$ if $Z_K = 0$, we obtain $R/I_K \cong H^0(\mathcal{O}_{Z_K})$ and $(R^1 \pi_* \mathcal{O}_Y)_0 \cong H^1(\mathcal{O}_{Z_K})$. Thus

$$\lambda(I_K) = h^0(\mathcal{O}_{Z_K}) \quad \text{and} \quad h^1(\mathcal{O}_{Z_K}) = p_g(X, 0).$$

By adjunction,

$$\omega_{Z_K} \cong (\omega_Y \otimes \mathcal{O}_Y(Z_K))|_{Z_K} \cong \mathcal{O}_{Z_K},$$

so Serre duality implies $h^0(\mathcal{O}_{Z_K}) = h^1(\mathcal{O}_{Z_K})$. Thus $\lambda(I_K) = p_g(X,0)$.

Let D be an exceptional cycle with $0 \le D \le Z_K$, and let $D' = Z_K - D$. Taking stalks at 0 of the direct images of

$$0 \to \mathcal{O}_Y(-Z_K) \to \mathcal{O}_Y(-D') \to \mathcal{O}_D(-D') \to 0$$

and using $R^1\pi_*\mathcal{O}_Y(-Z_K) = 0$, we obtain $H^0(\mathcal{O}_D(-D')) \cong I(D')/I_K$. Since $I_K \subset I(D')$ and $\lambda(I_K) = p_g(X,0)$,

$$h^0(\mathcal{O}_D(-D')) = p_g(X,0) - \lambda(I(D')).$$

By adjunction,

$$\omega_D \cong (\omega_Y \otimes \mathcal{O}_Y(D))|_D \cong \mathcal{O}_D(-D'),$$

so Serre duality implies

$$h^1(\mathcal{O}_D) = p_g(X,0) - \lambda(I(D')).$$

The kernel of the restriction $R \to H^0(\mathcal{O}_D)$ is $I(D)$, so $\lambda(I(D)) \le h^0(\mathcal{O}_D)$. Since $\chi(\mathcal{O}_D) = \chi(D)$,

$$\lambda(I(D)) \le h^0(\mathcal{O}_D) = \chi(D) + h^1(\mathcal{O}_D) = \chi(D) + p_g(X,0) - \lambda(I(D')).$$

For $D = 0$, both sides of the inequality in the statement are equal to $p_g(X,0)$. The lemma follows. □

3. An upper bound for the colength of a product of ideals

The main result of this section is Proposition 3.4, a lower bound for $4(\lambda(I)+\lambda(J)) - \lambda(IJ)$ for $\mathfrak{m}$-primary ideals $I \subset J$ of R. As in [KN17b, Lemma 26], the proof of Proposition 3.4 reduces to monomial ideals of $\mathbb{C}[x,y,z]$, for which we first prove Lemmas 3.1 and 3.2.

Lemma 3.1. *Let $I \subset J$ be monomial ideals of finite colength in $\mathbb{C}[x,y,z]$. Then*

$$4(\lambda(I)+\lambda(J)) - \lambda(IJ) \ge \lambda(I+(x)) + \lambda(I+(y)) + \lambda(J+(x)) + \lambda(J+(y)) + \left\lceil \sqrt{\lambda(I) - \lambda(J)} \right\rceil.$$

Proof. For a monomial ideal Q of finite colength in $\mathbb{C}[x,y,z]$ and $(i,j) \in \mathbb{N}^2$, let

$$\varphi_Q(i,j) = \min\{k \in \mathbb{N} \mid x^i y^j z^k \in Q\}.$$

Since Q contains a power of each of x,y,z, $\varphi_Q(i,j) < +\infty$ for any (i,j), and $\varphi_Q(i,j) = 0$ for $i+j \gg 0$. The monomials outside Q form a basis of $\mathbb{C}[x,y,z]/Q$, and $x^i y^j z^k \notin Q$ if and only if $k < \varphi_Q(i,j)$. The monomials outside $Q+(x)$ (resp. $Q+(y)$) are the monomials outside Q not divisible by x (resp. y). Thus

$$\lambda(Q) = \sum_{(i,j)\in\mathbb{N}^2} \varphi_Q(i,j), \qquad \lambda(Q+(x)) = \sum_{j\in\mathbb{N}} \varphi_Q(0,j), \qquad \lambda(Q+(y)) = \sum_{i\in\mathbb{N}} \varphi_Q(i,0). \tag{3.1}$$

Since $I \subset J$, $\varphi_J \le \varphi_I$. Let $\psi = \varphi_I + \varphi_J$ and $\delta = \varphi_I - \varphi_J$, two functions from $\mathbb{N}^2$ to $\mathbb{N}$ with finite support, and let

$$V = \sum_{(i,j)\in\mathbb{N}^2} \big(|\delta(i,j) - \delta(i+1,j)| + |\delta(i,j) - \delta(i,j+1)|\big).$$

Step 1. We prove

$$\lambda(IJ) \le 4(\lambda(I)+\lambda(J)) - \sum_{Q=I,J} \big(\lambda(Q+(x)) + \lambda(Q+(y))\big) - \frac{V}{2}. \tag{3.2}$$

For any $(i,j),(i',j') \in \mathbb{N}^2$, $x^{i+i'}y^{j+j'}z^{\varphi_I(i,j)+\varphi_J(i',j')} \in IJ$, so

$$\varphi_{IJ}(i+i',j+j') \le \varphi_I(i,j) + \varphi_J(i',j'). \tag{3.3}$$

For $(i,j) \in \mathbb{N}^2$, let

$$\eta(i,j) = \psi(i,j) + \psi(i+1,j+1) - \psi(i+1,j) - \psi(i,j+1).$$

We apply (3.3) with $((i,j),(i',j'))$ as follows, and use $\min\{\alpha,\beta\} = (\alpha+\beta-|\alpha-\beta|)/2$ for $\alpha,\beta \in \mathbb{R}$.

(1) $((i,j),(i,j))$: $\varphi_{IJ}(2i,2j) \le \psi(i,j)$.
(2) $((i,j),(i+1,j))$ and $((i+1,j),(i,j))$:

$$\begin{aligned}\varphi_{IJ}(2i+1,2j) &\le \min\{\varphi_I(i,j)+\varphi_J(i+1,j), \varphi_I(i+1,j)+\varphi_J(i,j)\}\\ &= \frac{\psi(i,j)+\psi(i+1,j)-|\delta(i,j)-\delta(i+1,j)|}{2},\end{aligned}$$

since $\alpha = \varphi_I(i,j)+\varphi_J(i+1,j)$ and $\beta = \varphi_I(i+1,j)+\varphi_J(i,j)$ satisfy

$$\alpha+\beta = \psi(i,j)+\psi(i+1,j) \qquad \text{and} \qquad \alpha-\beta = \delta(i,j)-\delta(i+1,j).$$

(3) $((i,j),(i,j+1))$ and $((i,j+1),(i,j))$:

$$\varphi_{IJ}(2i,2j+1) \le \frac{\psi(i,j)+\psi(i,j+1)-|\delta(i,j)-\delta(i,j+1)|}{2}.$$

(4) $((i,j),(i+1,j+1))$ and $((i+1,j+1),(i,j))$, resp. $((i+1,j),(i,j+1))$ and $((i,j+1),(i+1,j))$:

$$\varphi_{IJ}(2i+1,2j+1) \le \frac{\psi(i,j)+\psi(i+1,j+1)}{2},$$
$$\varphi_{IJ}(2i+1,2j+1) \le \frac{\psi(i+1,j)+\psi(i,j+1)}{2}.$$

Thus

$$\varphi_{IJ}(2i+1,2j+1) \le \frac{\psi(i,j)+\psi(i+1,j+1)+\psi(i+1,j)+\psi(i,j+1)-|\eta(i,j)|}{4}.$$

Let

$$\Psi = \sum_{(i,j)\in\mathbb{N}^2} \psi(i,j) = \lambda(I)+\lambda(J).$$

Since ψ has finite support,

$$\sum_{(i,j)\in\mathbb{N}^2} \psi(i+1,j) = \Psi - \sum_{j\in\mathbb{N}} \psi(0,j), \qquad \sum_{(i,j)\in\mathbb{N}^2} \psi(i,j+1) = \Psi - \sum_{i\in\mathbb{N}} \psi(i,0),$$

$$\sum_{(i,j)\in\mathbb{N}^2} \psi(i+1,j+1) = \Psi - \sum_{j\in\mathbb{N}} \psi(0,j) - \sum_{i\in\mathbb{N}} \psi(i,0) + \psi(0,0).$$

Thus $\sum_{(i,j)\in\mathbb{N}^2} \eta(i,j) = \psi(0,0)$, so

$$\sum_{(i,j)\in\mathbb{N}^2} |\eta(i,j)| \ge \psi(0,0).$$

Any element of $\mathbb{N}^2$ is of exactly one of the forms $(2i,2j)$, $(2i+1,2j)$, $(2i,2j+1)$, $(2i+1,2j+1)$ with $(i,j)\in\mathbb{N}^2$, and (i,j) is unique. Since $\lambda(IJ) = \sum_{(i,j)\in\mathbb{N}^2} \varphi_{IJ}(i,j)$ by (3.1), adding (1)–(4) over $(i,j)\in\mathbb{N}^2$ implies

$$\begin{aligned}\lambda(IJ) &\le 4\Psi - \sum_{j\in\mathbb{N}} \psi(0,j) - \sum_{i\in\mathbb{N}} \psi(i,0) + \frac{1}{4}\Big(\psi(0,0) - \sum_{(i,j)\in\mathbb{N}^2} |\eta(i,j)|\Big) - \frac{V}{2}\\ &\le 4\Psi - \sum_{j\in\mathbb{N}} \psi(0,j) - \sum_{i\in\mathbb{N}} \psi(i,0) - \frac{V}{2}.\end{aligned}$$

By (3.1), $\sum_{j\in\mathbb{N}} \psi(0,j) = \lambda(I+(x)) + \lambda(J+(x))$ and $\sum_{i\in\mathbb{N}} \psi(i,0) = \lambda(I+(y)) + \lambda(J+(y))$, which implies (3.2).

Step 2. We prove

$$V \ge 2\sqrt{\lambda(I)-\lambda(J)}. \tag{3.4}$$

For any integer $k \ge 1$, let

$$\Lambda_k = \{(i,j)\in\mathbb{N}^2 \mid \delta(i,j) \ge k\},$$

and let $a_k = \#p_1(\Lambda_k)$ and $b_k = \#p_2(\Lambda_k)$, where $p_1, p_2 : \mathbb{N}^2 \to \mathbb{N}$ are the projections to the first and the second coordinate. Λ_k is finite since δ has finite support. Since δ takes values in $\mathbb{N}$, for any $(i,j),(i',j') \in \mathbb{N}^2$,

$$|\delta(i,j) - \delta(i',j')| = \#\{k \geq 1 \mid \text{exactly one of } (i,j),(i',j') \text{ belongs to } \Lambda_k\}.$$

Thus $V = \sum_{k\geq 1} V_k$, where

$$\begin{aligned} V_k = {} & \#\{(i,j) \in \mathbb{N}^2 \mid \text{exactly one of } (i,j),(i+1,j) \text{ belongs to } \Lambda_k\} \\ & + \#\{(i,j) \in \mathbb{N}^2 \mid \text{exactly one of } (i,j),(i,j+1) \text{ belongs to } \Lambda_k\}. \end{aligned}$$

Fix $k \geq 1$ and $j \in p_2(\Lambda_k)$. Since Λ_k is finite, there exists $i_0 \in \mathbb{N}$ such that $(i_0, j) \in \Lambda_k$ and $(i_0+1, j) \notin \Lambda_k$. Thus the first set in V_k has at least b_k elements. Similarly, for $i \in p_1(\Lambda_k)$, there exists $j_0 \in \mathbb{N}$ such that $(i, j_0) \in \Lambda_k$ and $(i, j_0+1) \notin \Lambda_k$, so the second set in V_k has at least a_k elements. Since $\Lambda_k \subset p_1(\Lambda_k) \times p_2(\Lambda_k)$,

$$\#\Lambda_k \leq a_k b_k \leq \Big(\frac{a_k + b_k}{2}\Big)^2 \leq \Big(\frac{V_k}{2}\Big)^2.$$

Any $(i,j) \in \mathbb{N}^2$ belongs to Λ_k for exactly $\delta(i,j)$ values of $k \geq 1$, so by (3.1)

$$\sum_{k\geq 1} \#\Lambda_k = \sum_{(i,j)\in\mathbb{N}^2} \delta(i,j) = \lambda(I) - \lambda(J).$$

Therefore

$$V \geq 2\sum_{k\geq 1} \sqrt{\#\Lambda_k} \geq 2\sqrt{\sum_{k\geq 1} \#\Lambda_k} = 2\sqrt{\lambda(I) - \lambda(J)}.$$

By (3.2) and (3.4),

$$4(\lambda(I) + \lambda(J)) - \lambda(IJ) - \sum_{Q=I,J} \big(\lambda(Q + (x)) + \lambda(Q + (y))\big) \geq \frac{V}{2} \geq \sqrt{\lambda(I) - \lambda(J)}.$$

The left-hand side is an integer, so it is at least $\lceil \sqrt{\lambda(I) - \lambda(J)} \rceil$. The lemma follows. □

Lemma 3.2. *Let I and J be monomial ideals of finite colength in $\mathbb{C}[u,v]$. Then*

$$2(\lambda(I) + \lambda(J)) - \lambda(IJ) \geq \lambda(I + (u)).$$

Proof. For a monomial ideal Q of finite colength in $\mathbb{C}[u,v]$ and $i \in \mathbb{N}$, let

$$\varphi_Q(i) = \min\{k \in \mathbb{N} \mid u^i v^k \in Q\}.$$

Then $\varphi_Q(i) < +\infty$, and $\varphi_Q(i) = 0$ for $i \gg 0$. Since the monomials outside Q form a basis of $\mathbb{C}[u,v]/Q$,

$$\lambda(Q) = \sum_{i\in\mathbb{N}} \varphi_Q(i), \qquad \lambda(Q + (u)) = \varphi_Q(0).$$

Let $\psi = \varphi_I + \varphi_J$, $\delta = \varphi_I - \varphi_J$, and $\Psi = \sum_{i\in\mathbb{N}} \psi(i) = \lambda(I) + \lambda(J)$. Since $u^{i+i'} v^{\varphi_I(i)+\varphi_J(i')} \in IJ$ for any $i, i' \in \mathbb{N}$,

$$\begin{aligned} \varphi_{IJ}(2i) &\leq \psi(i), \\ \varphi_{IJ}(2i+1) &\leq \min\{\varphi_I(i) + \varphi_J(i+1), \varphi_I(i+1) + \varphi_J(i)\} \\ &= \frac{\psi(i) + \psi(i+1) - |\delta(i) - \delta(i+1)|}{2}. \end{aligned}$$

Since $\sum_{i\in\mathbb{N}} \psi(i+1) = \Psi - \psi(0)$, adding the bounds for $\varphi_{IJ}(2i)$ and $\varphi_{IJ}(2i+1)$ over $i \in \mathbb{N}$ implies

$$\lambda(IJ) \leq 2\Psi - \frac{\psi(0)}{2} - \frac{1}{2}\sum_{i\in\mathbb{N}} |\delta(i) - \delta(i+1)|.$$

Since δ has finite support, $\sum_{i\in\mathbb{N}}(\delta(i)-\delta(i+1))=\delta(0)$, so $\sum_{i\in\mathbb{N}}|\delta(i)-\delta(i+1)|\geq\delta(0)$. As $\psi(0)+\delta(0)=2\varphi_I(0)=2\lambda(I+(u))$,

$$\lambda(IJ)\leq 2(\lambda(I)+\lambda(J))-\lambda(I+(u)).$$

The lemma follows. □

Definition 3.3. Notation as in Definition 2.1. For an ideal $Q\subset R$ with $\lambda(Q)<+\infty$, let

$$c(Q)=\min\{\lambda(Q+tR)\mid t\in\mathbb{C}x+\mathbb{C}y+\mathbb{C}z,\ t\neq 0\},$$

where tR is the ideal of R generated by the image of t. If moreover $Q\neq R$, the *Loewy length* of R/Q is $N_Q=\min\{j\geq 1\mid\mathfrak{m}^j\subset Q\}$.

Proposition 3.4. *Notation as in Definition 3.3. Let $I\subset J$ be $\mathfrak{m}$-primary ideals of R. Then*

$$4(\lambda(I)+\lambda(J))-\lambda(IJ)\geq c(IJ)+N_I+\left\lceil\sqrt{\lambda(I)-\lambda(J)}\right\rceil.$$

Proof. Let $\widetilde{I}\subset\widetilde{J}\subset S$ be the inverse images of I and J. Then the inverse image of IJ is $\widetilde{IJ}=\widetilde{I}\widetilde{J}+\mathfrak{a}$, and

$$\lambda(I)=\dim_{\mathbb{C}}S/\widetilde{I},\qquad\lambda(J)=\dim_{\mathbb{C}}S/\widetilde{J},\qquad\lambda(IJ)=\dim_{\mathbb{C}}S/\widetilde{IJ}.$$

Step 1. For an $\mathfrak{n}$-primary ideal $Q\subset S$, let $Q^*\subset\mathbb{C}[x,y,z]$ be the ideal generated by the lowest-degree homogeneous terms of the non-zero elements of Q. Since $\mathrm{gr}_{\mathfrak{n}}S\cong\mathbb{C}[x,y,z]$, we have an isomorphism of graded rings

$$\mathrm{gr}_{\mathfrak{n}}(S/Q)\cong\mathbb{C}[x,y,z]/Q^*$$

[Eis95, Section 5.4 and Exercise 5.3], and the degree-j part of $\mathrm{gr}_{\mathfrak{n}}(S/Q)$ is $(\mathfrak{n}^j+Q)/(\mathfrak{n}^{j+1}+Q)$. Since Q contains a power of $\mathfrak{n}$,

$$\dim_{\mathbb{C}}S/Q=\dim_{\mathbb{C}}\mathbb{C}[x,y,z]/Q^*. \tag{3.5}$$

By Nakayama's lemma, $(\mathfrak{n}^j+Q)/(\mathfrak{n}^{j+1}+Q)=0$ if and only if $\mathfrak{n}^j\subset Q$. Thus

$$\max\{j\in\mathbb{N}\mid(\mathbb{C}[x,y,z]/Q^*)_j\neq 0\}=\min\{j\mid\mathfrak{n}^j\subset Q\}-1.$$

We write $I^*=\widetilde{I}^*$, $J^*=\widetilde{J}^*$, and $(IJ)^*=\widetilde{IJ}^*$. Since $\widetilde{I}\supset\mathfrak{a}$, $\mathfrak{n}^j\subset\widetilde{I}$ if and only if $\mathfrak{m}^j\subset I$, so $\min\{j\mid\mathfrak{n}^j\subset\widetilde{I}\}=N_I$ and

$$\max\{j\in\mathbb{N}\mid(\mathbb{C}[x,y,z]/I^*)_j\neq 0\}=N_I-1. \tag{3.6}$$

Since $\widetilde{I}\subset\widetilde{J}$, $I^*\subset J^*$. Since the lowest-degree term of a product of two non-zero elements of S is the product of their lowest-degree terms,

$$I^*J^*\subset(IJ)^*.$$

Step 2. $GL_3(\mathbb{C})$ acts on $\mathbb{C}x+\mathbb{C}y+\mathbb{C}z$, hence on $\mathbb{C}[x,y,z]$ and on S by $\mathbb{C}$-algebra automorphisms which preserve the degree of homogeneous forms. Thus $(\sigma\cdot Q)^*=\sigma\cdot Q^*$ for any $\sigma\in GL_3(\mathbb{C})$ and any $\mathfrak{n}$-primary ideal $Q\subset S$.

Since the (N_I-1)-th powers of linear forms span the space $\mathbb{C}[x,y,z]_{N_I-1}$ of forms of degree N_I-1, by (3.6) there exists a non-zero linear form t with $t^{N_I-1}\notin I^*$. We fix $g\in GL_3(\mathbb{C})$ with $g\cdot t=z$, so that $z^{N_I-1}\notin g\cdot I^*$.

Fix a monomial order on $\mathbb{C}[x,y,z]$ which refines the total degree and satisfies $x>y>z$, and let

$$I'=\mathrm{in}(g\cdot I^*),\qquad J'=\mathrm{in}(g\cdot J^*),\qquad(IJ)'=\mathrm{in}\big(g\cdot(IJ)^*\big).$$

For a homogeneous ideal $Q\subset\mathbb{C}[x,y,z]$ and $j\in\mathbb{N}$,

$$\dim_{\mathbb{C}}\big(\mathbb{C}[x,y,z]/\mathrm{in}(Q)\big)_j=\dim_{\mathbb{C}}\big(\mathbb{C}[x,y,z]/Q\big)_j=\dim_{\mathbb{C}}\big(\mathbb{C}[x,y,z]/g\cdot Q\big)_j,$$

the first equality by [Eis95, Theorem 15.26] and the second since g preserves the degree of homogeneous forms. By (3.5),

$$\lambda(I')=\lambda(I),\qquad\lambda(J')=\lambda(J),\qquad\lambda((IJ)')=\lambda(IJ). \tag{3.7}$$

Since g and in preserve inclusions, and the initial monomial of a product of two non-zero polynomials is the product of their initial monomials, **Step 1** implies

$$I' \subset J', \qquad I'J' \subset (IJ)'. \tag{3.8}$$

Step 3. We prove

$$\lambda(I' + (x,y)) = N_I. \tag{3.9}$$

Since $x > z$ and $y > z$, $x^i y^j z^k > z^{i+j+k}$ whenever $i + j > 0$, so z^{N_I-1} is the smallest monomial of degree $N_I - 1$. Let $\mathfrak{b} = g \cdot I^*$. If $z^{N_I-1} \in \mathrm{in}(\mathfrak{b})$, then $z^{N_I-1} = \mathrm{in}(b)$ for some $b \in \mathfrak{b}$. Since the order refines the degree and $\mathfrak{b}$ is homogeneous, we may replace b by the homogeneous component of b of degree $N_I - 1$, and then each monomial of b is at most z^{N_I-1}, so b is a non-zero scalar multiple of z^{N_I-1}. Thus $z^{N_I-1} \in \mathrm{in}(\mathfrak{b})$ if and only if $z^{N_I-1} \in \mathfrak{b}$, and so $z^{N_I-1} \notin I'$. By (3.6), $\mathfrak{b}$ contains any form of degree N_I, so $z^{N_I} \in I'$. Since I' is an ideal, $z^k \notin I'$ if and only if $k < N_I$, so the monomials outside $I' + (x,y)$ are the z^k with $0 \le k < N_I$. This implies (3.9).

Step 4. Let $\ell \in \{x, y\}$. We prove

$$\lambda(I'J' + (\ell)) \ge c(IJ). \tag{3.10}$$

By (3.8) and since (ℓ) is generated by a monomial,

$$I'J' + (\ell) \subset (IJ)' + (\ell) \subset \mathrm{in}\big(g \cdot (IJ)^* + (\ell)\big). \tag{3.11}$$

By **Step 2**, $g \cdot (IJ)^* = (g \cdot \widetilde{IJ})^*$, and since ℓ is homogeneous,

$$(g \cdot \widetilde{IJ})^* + (\ell) \subset (g \cdot \widetilde{IJ} + \ell S)^*. \tag{3.12}$$

By [Eis95, Theorem 15.26] for the homogeneous ideal $g \cdot (IJ)^* + (\ell)$, (3.5) for the $\mathfrak{n}$-primary ideal $g \cdot \widetilde{IJ} + \ell S$, and the inclusions (3.11) and (3.12),

$$\lambda(I'J' + (\ell)) \ge \dim_{\mathbb{C}} \mathbb{C}[x,y,z]/\big(g \cdot (IJ)^* + (\ell)\big) \ge \dim_{\mathbb{C}} S/(g \cdot \widetilde{IJ} + \ell S).$$

The automorphism g^{-1} of S maps $g \cdot \widetilde{IJ} + \ell S$ onto $\widetilde{IJ} + g^{-1}(\ell)S$, which is the inverse image of $IJ + g^{-1}(\ell)R$. Since $g^{-1}(\ell)$ is a non-zero linear form,

$$\dim_{\mathbb{C}} S/(g \cdot \widetilde{IJ} + \ell S) = \lambda\big(IJ + g^{-1}(\ell)R\big) \ge c(IJ),$$

which implies (3.10).

Step 5. Let $\{\ell, w\} = \{x, y\}$. The images of $I' \subset J'$ in $\mathbb{C}[x,y,z]/(\ell) \cong \mathbb{C}[w,z]$ are monomial ideals of finite colength, and the product of the images of I' and J' is the image of $I'J'$. The colengths of the images of I', J', $I'J'$ are $\lambda(I' + (\ell))$, $\lambda(J' + (\ell))$, $\lambda(I'J' + (\ell))$, and the colength of the image of $I' + (w)$ is $\lambda(I' + (x,y)) = N_I$ by (3.9). By Lemma 3.2 with $(u,v) = (w,z)$,

$$2\big(\lambda(I' + (\ell)) + \lambda(J' + (\ell))\big) - \lambda(I'J' + (\ell)) \ge N_I. \tag{3.13}$$

Adding (3.13) for $\ell = x$ and for $\ell = y$ and using (3.10) for $\ell = x$ and $\ell = y$,

$$\begin{aligned} 2\sum_{Q=I',J'} \big(\lambda(Q + (x)) + \lambda(Q + (y))\big) &\ge 2N_I + \lambda(I'J' + (x)) + \lambda(I'J' + (y)) \\ &\ge 2N_I + 2c(IJ), \end{aligned}$$

thus

$$\sum_{Q=I',J'} \big(\lambda(Q + (x)) + \lambda(Q + (y))\big) \ge N_I + c(IJ).$$

By (3.7) and (3.8), $\lambda(IJ) = \lambda((IJ)') \le \lambda(I'J')$. By Lemma 3.1 for $I' \subset J'$ and (3.7),

$$\begin{aligned} 4(\lambda(I) + \lambda(J)) - \lambda(IJ) &\ge 4(\lambda(I') + \lambda(J')) - \lambda(I'J') \\ &\ge \sum_{Q=I',J'} \big(\lambda(Q + (x)) + \lambda(Q + (y))\big) + \left\lceil \sqrt{\lambda(I') - \lambda(J')} \right\rceil \\ &\ge c(IJ) + N_I + \left\lceil \sqrt{\lambda(I) - \lambda(J)} \right\rceil. \end{aligned}$$

The proposition follows. □

4. Linear sections and order of the adjoint ideal

The main result of this section is Proposition 4.2, a lower bound for the order of the elements of S whose images in R belong to the adjoint ideal I_K.

Lemma 4.1. *Notation and conditions as in Set-up 2.2. Then*

$$c(I_K) = h^1(\mathcal{O}_M) \leq p_g(X, 0),$$

where c is as in Definition 3.3.

Proof. Let $v_i(0) = +\infty$ for any i. Let $D = \sum_{i=1}^n d_i E_i$ be an exceptional cycle, and let V be an open neighborhood of 0 in X. A section of $\mathcal{O}_Y(-D)$ over $\pi^{-1}(V)$ is holomorphic on $\pi^{-1}(V)\backslash E \cong V\backslash\{0\}$, so it is the pullback of a holomorphic function on V by Hartogs extension on the normal surface X. Since the divisor of $\pi^* u$ has non-negative coefficients along the non-exceptional prime divisors for any $u \in R\backslash\{0\}$, we obtain

$$(\pi_* \mathcal{O}_Y(-D))_0 = \{u \in R \mid v_i(u) \geq d_i \text{ for any } i\}. \tag{4.1}$$

Since the images of x, y, z generate $\mathfrak{m}$, $m_i = \min\{v_i(x), v_i(y), v_i(z)\}$ for any i. Thus a general $t \in \mathbb{C}x + \mathbb{C}y + \mathbb{C}z$ satisfies $v_i(t) = m_i$ for any i, where $v_i(t)$ denotes v_i of the image of t in R. We fix such a t. By (4.1) for $D = Z_K$,

$$I_K = \{u \in R \mid v_i(u) \geq \operatorname{mult}_{E_i} Z_K \text{ for any } i\}.$$

Since $t \in \mathfrak{m}$, $(I_K : \mathfrak{m}) \subset (I_K : t)$. If $u \in (I_K : t)$, then $v_i(u) + m_i = v_i(ut) \geq \operatorname{mult}_{E_i} Z_K$ for any i. If $v_i(u) \geq \operatorname{mult}_{E_i} Z_K - m_i$ for any i, then $v_i(ua) \geq v_i(u) + m_i \geq \operatorname{mult}_{E_i} Z_K$ for any $a \in \mathfrak{m}$ and any i, so $u \in (I_K : \mathfrak{m})$. By (4.1) for $D = Z_K - M$, we obtain

$$(I_K : \mathfrak{m}) = (I_K : t) = \{u \in R \mid v_i(u) \geq \operatorname{mult}_{E_i} Z_K - m_i \text{ for any } i\} = (\pi_* \mathcal{O}_Y(M - Z_K))_0. \tag{4.2}$$

Since $m_i \geq 1$ for any i, $M \geq 0$ is a divisor on Y with support E, hence a compact Cohen–Macaulay curve. Since $\omega_Y \cong \mathcal{O}_Y(-Z_K)$, adjunction on the smooth surface Y gives $\omega_M \cong \mathcal{O}_Y(M - Z_K)|_M$, so

$$0 \to \mathcal{O}_Y(-Z_K) \to \mathcal{O}_Y(M - Z_K) \to \omega_M \to 0$$

is exact. By Grauert–Riemenschneider vanishing, $R^1\pi_* \mathcal{O}_Y(-Z_K) \cong R^1\pi_* \omega_Y = 0$. Taking the stalks at 0 of the direct images under π, by (4.2) we obtain $(I_K : \mathfrak{m})/I_K \cong H^0(M, \omega_M)$. By Serre duality on M, $\dim_{\mathbb{C}} H^0(M, \omega_M) = h^1(\mathcal{O}_M)$. Thus

$$\dim_{\mathbb{C}}(I_K : \mathfrak{m})/I_K = h^1(\mathcal{O}_M). \tag{4.3}$$

By Lemma 2.4, $\dim_{\mathbb{C}} R/I_K = \lambda(I_K) = p_g(X, 0) < +\infty$. Let $t' \in \mathfrak{m}$. The kernel and the cokernel of multiplication by t' on R/I_K are $(I_K : t')/I_K$ and $R/(I_K + t'R)$. Since $\dim_{\mathbb{C}} R/I_K < +\infty$, $\dim_{\mathbb{C}}(I_K : t')/I_K = \dim_{\mathbb{C}} R/(I_K + t'R)$. Since $(I_K : \mathfrak{m}) \subset (I_K : t')$, by (4.3) we have

$$\lambda(I_K + t'R) = \dim_{\mathbb{C}}(I_K : t')/I_K \geq h^1(\mathcal{O}_M),$$

and equality holds for $t' = t$ by (4.2). Since the image of any non-zero $t' \in \mathbb{C}x + \mathbb{C}y + \mathbb{C}z$ in R belongs to $\mathfrak{m}$, $c(I_K) = h^1(\mathcal{O}_M)$. Finally, by (4.3),

$$h^1(\mathcal{O}_M) = \dim_{\mathbb{C}}(I_K : \mathfrak{m})/I_K \leq \dim_{\mathbb{C}} R/I_K = p_g(X, 0).$$

The lemma follows. □

Proposition 4.2. *Notation and conditions as in Set-up 2.2. Then the inverse image of I_K in S is contained in $\mathfrak{n}^{\nu-2}$. In particular,*

$$p_g(X, 0) \geq \binom{\nu}{3}.$$

Proof. Let $\widetilde{I}_K \subset S$ be the inverse image of I_K. Possibly shrinking X to a neighborhood of 0, we may assume that $X = \{f = 0\} \subset U$ for an open neighborhood U of 0 in $\mathbb{C}^3$ on which f converges.

Let $\eta = \mathrm{Res}(dx \wedge dy \wedge dz/f)$ be the Poincaré residue. Then η is a holomorphic 2-form without zeros on $X\backslash\{0\}$, and η generates ω_X. The divisor of the meromorphic 2-form $\pi^*\eta$ on Y is a canonical divisor supported on E, hence equal to $-Z_K$ by the definition of Z_K.

Let $\beta : U_1 \to U$ be the blow-up of 0 with exceptional divisor F_1, and let $\sigma : W \to U$ be an embedded resolution of X, so that W and $X' = \sigma_*^{-1}X$ are smooth, such that σ is an isomorphism over $U\backslash\{0\}$ and factors as $\sigma = \beta \circ \tau$ for a morphism $\tau : W \to U_1$. Let $\sigma' = \sigma|_{X'} : X' \to X$, and let $F = \tau_*^{-1}F_1$. Since σ' is a resolution of X and π is the minimal resolution, $\sigma' = \pi \circ \rho$ for a morphism $\rho : X' \to Y$.

Let $g \in I_K\backslash\{0\}$. Since π^*g is a section of $\mathcal{O}_Y(-Z_K)$ and the divisor of $\pi^*\eta$ is $-Z_K$, $\pi^*(g\eta)$ is a holomorphic 2-form on Y. Thus $\sigma'^*(g\eta) = \rho^*\pi^*(g\eta)$ is a holomorphic 2-form on X'. By adjunction for the smooth divisor $X' \subset W$, the residue map fits into an exact sequence

$$0 \to \omega_W \to \omega_W(X') \to \omega_{X'} \to 0.$$

By Grauert–Riemenschneider vanishing, $R^1\sigma_*\omega_W = 0$, so the residue map $\sigma_*\omega_W(X') \to \sigma'_*\omega_{X'}$ is surjective. Thus, possibly shrinking U to a neighborhood of 0, there exists a section Φ of $\omega_W(X')$ over W whose residue on X' is $\sigma'^*(g\eta)$.

Since Φ has at most a simple pole along X' and σ is an isomorphism over $U\backslash\{0\}$, there exists a holomorphic function $\widetilde{g}$ on $U\backslash\{0\}$ such that

$$\Phi = \sigma^*\Big(\widetilde{g}\frac{dx \wedge dy \wedge dz}{f}\Big) \quad \text{over } U\backslash\{0\}.$$

Since $\mathrm{codim}_U\{0\} = 3$, $\widetilde{g}$ extends to a holomorphic function on U by Hartogs extension, and we denote the germ of $\widetilde{g}$ at 0 again by $\widetilde{g} \in S$. On $X\backslash\{0\}$ we have

$$(\widetilde{g}|_X)\eta = \mathrm{Res}\left(\widetilde{g}\frac{dx \wedge dy \wedge dz}{f}\right) = g\eta.$$

Since η has no zeros on $X\backslash\{0\}$, the image of $\widetilde{g}$ in R is g. Thus $\widetilde{g} \neq 0$.

Since $\sigma = \beta \circ \tau$ and $F = \tau_*^{-1}F_1$,

$$\mathrm{mult}_F\, \sigma^*\widetilde{g} = \mathrm{mult}_0\, \widetilde{g}, \quad \mathrm{mult}_F\, \mathrm{div}(\sigma^*(dx \wedge dy \wedge dz)) = 2, \quad \mathrm{mult}_F\, \sigma^*f = \nu.$$

Since $F \neq X'$ and Φ is a section of $\omega_W(X')$, Φ has no pole along F. Thus

$$0 \leq \mathrm{mult}_F\, \mathrm{div}(\Phi) = \mathrm{mult}_0\, \widetilde{g} + 2 - \nu,$$

so $\widetilde{g} \in \mathfrak{n}^{\nu-2}$. Any other element of S whose image in R is g is $\widetilde{g} + fu$ for some $u \in S$, and $f \in \mathfrak{n}^\nu \subset \mathfrak{n}^{\nu-2}$. Since $g \in I_K\backslash\{0\}$ is arbitrary and $f \in \mathfrak{n}^{\nu-2}$, we obtain

$$\widetilde{I}_K \subset \mathfrak{n}^{\nu-2}.$$

By Lemma 2.4,

$$p_g(X,0) = \lambda(I_K) = \dim_{\mathbb{C}} S/\widetilde{I}_K \geq \dim_{\mathbb{C}} S/\mathfrak{n}^{\nu-2} = \binom{\nu}{3},$$

where the last equality counts the monomials in x, y, z of degree at most $\nu - 3$. For $\nu \geq 3$, where $(X,0)$, being Gorenstein and not a rational double point, satisfies the assumptions of [KN17b, Section 4], the inequality $p_g(X,0) \geq \binom{\nu}{3}$ also follows from the argument for strict complete intersections in [KN17b, Section 4], namely the semicontinuity of the geometric genus together with [KN17b, (24)] for a single defining equation of degree ν. For $\nu \leq 2$ the inequality is trivial. □

5. Lower bounds for $q - b_1(E)$

The main result of this section is Proposition 5.4.

Lemma 5.1. *Let C be a compact reduced curve, possibly empty or disconnected. Then*

$$2\chi(\mathcal{O}_C) \le \chi_{\mathrm{top}}(C),$$

where χ_{top} is the topological Euler characteristic. If moreover C is connected, then

$$\dim_{\mathbb{C}} H^1(C, \mathbb{C}) \le 2h^1(\mathcal{O}_C).$$

Proof. Let $C_1, \dots, C_N$ be the irreducible components of C, let $\widetilde{C}_j \to C_j$ be the normalization of C_j, let g_j be the genus of $\widetilde{C}_j$, and let $\gamma : \widetilde{C} = \bigsqcup_{j=1}^N \widetilde{C}_j \to C$ be the normalization of C. For $P \in \operatorname{Sing} C$, let $r_P = \#\gamma^{-1}(P)$, and let

$$\delta_P = \dim_{\mathbb{C}} \overline{\mathcal{O}}_{C,P}/\mathcal{O}_{C,P},$$

where $\overline{\mathcal{O}}_{C,P}$ is the integral closure of $\mathcal{O}_{C,P}$. All sums over P below run over $P \in \operatorname{Sing} C$, and empty sums are 0. The normalization exact sequence implies

$$\chi(\mathcal{O}_C) = \sum_{j=1}^N (1 - g_j) - \sum_P \delta_P.$$

Topologically, C is the quotient space of $\widetilde{C}$ in which, for any $P \in \operatorname{Sing} C$, the r_P points of $\gamma^{-1}(P)$ are identified to one point. Thus

$$\chi_{\mathrm{top}}(C) = \sum_{j=1}^N (2 - 2g_j) - \sum_P (r_P - 1).$$

Since $\delta_P \ge r_P - 1 \ge 0$ for any P,

$$2\chi(\mathcal{O}_C) = \sum_{j=1}^N (2 - 2g_j) - 2\sum_P \delta_P \le \sum_{j=1}^N (2 - 2g_j) - \sum_P (r_P - 1) = \chi_{\mathrm{top}}(C).$$

Assume that C is connected. Then $h^0(\mathcal{O}_C) = 1$, $\dim_{\mathbb{C}} H^0(C, \mathbb{C}) = 1$, and $\dim_{\mathbb{C}} H^2(C, \mathbb{C}) = N$. Thus

$$h^1(\mathcal{O}_C) = \sum_{j=1}^N g_j + \sum_P \delta_P - N + 1 \quad \text{and} \quad \dim_{\mathbb{C}} H^1(C, \mathbb{C}) = 2\sum_{j=1}^N g_j + \sum_P (r_P - 1) - N + 1.$$

Let Γ be the incidence graph of C: the vertices of Γ are $C_1, \dots, C_N$ and the points of $\operatorname{Sing} C$, and the edges of Γ are the points of $\gamma^{-1}(\operatorname{Sing} C)$, the edge given by a point of $\widetilde{C}_j \cap \gamma^{-1}(P)$ joining P and C_j. Since C is connected, Γ is connected. Since Γ has $N + \#\operatorname{Sing} C$ vertices and $\sum_P r_P$ edges, the cycle rank of Γ is

$$\sum_P (r_P - 1) - N + 1 \ge 0.$$

Thus

$$2h^1(\mathcal{O}_C) - \dim_{\mathbb{C}} H^1(C, \mathbb{C}) = 2\sum_P \big(\delta_P - (r_P - 1)\big) + \sum_P (r_P - 1) - N + 1 \ge 0.$$

□

Lemma 5.2. *Let f, ν, π, and M be as in Set-up 2.2. Then $-M^2 \le \nu$.*

Proof. Since the zero set of $\mathfrak{m}$ is $\{0\}$, the definition of M implies

$$\mathfrak{m}\mathcal{O}_Y = \mathcal{O}_Y(-M) \cdot \mathcal{J}$$

for an ideal $\mathcal{J} \subset \mathcal{O}_Y$ whose cosupport is a finite subset of E. Let $\rho : Y' \to Y$ be a composition of point blow-ups such that $\mathcal{J}\mathcal{O}_{Y'}$ is invertible, and write $\mathcal{J}\mathcal{O}_{Y'} = \mathcal{O}_{Y'}(-F)$, where $F \geq 0$ is ρ-exceptional. Then $\mathfrak{m}\mathcal{O}_{Y'} = \mathcal{O}_{Y'}(-M')$ with

$$M' = \rho^* M + F.$$

Since F is ρ-exceptional, $\rho^* M \cdot F = 0$, and the intersection form on the ρ-exceptional curves is negative definite. Thus

$$-M'^2 = -M^2 - F^2 \geq -M^2.$$

Since $\pi \circ \rho$ is a resolution of X and $\mathfrak{m}\mathcal{O}_{Y'} = \mathcal{O}_{Y'}(-M')$, the multiplicity of R is $-M'^2$ by [Wag70, Theorem 2.7]. The multiplicity of the hypersurface ring $R = S/(f)$ is ν. Thus $-M^2 \leq -M'^2 = \nu$. □

Definition 5.3. Let π, $E_1, \dots, E_n$, and Z_K be as in Set-up 2.2, and let

$$q = n + \max\{(Z_K - 2D)^2 : D \text{ is an exceptional cycle}\}.$$

The maximum exists since the intersection matrix of $E_1, \dots, E_n$ is negative definite, so that the values $(Z_K - 2D)^2$ are integers ≤ 0.

Proposition 5.4. *Let f, ν, π, E, $b_1(E)$, Z_K, and M be as in Set-up 2.2 and q as in Definition 5.3. Then the following hold.*

(1) $q \geq b_1(E) - 1 + Z_K \cdot E$.
(2) $q + 2h^1(\mathcal{O}_M) - b_1(E) \geq 1 - \nu$.

Proof. The proof of (1) follows [KN17b, (16)–(19)].

We prove (1). Write $Z_K = \sum_i a_i E_i$ with $a_i \in \mathbb{Z}$. Let E' be the sum of the E_i such that a_i is odd, and let $E'' = E - E'$. The coefficients of $Z_K - E'$ and of $Z_K - 2E + E'$ along E_i are $a_i - \operatorname{mult}_{E_i} E'$ and $a_i + \operatorname{mult}_{E_i} E' - 2$, which are even. Thus

$$E' = Z_K - 2D' \quad \text{and} \quad E + E'' = Z_K - 2(D' - E''), \quad \text{where } D' = \frac{Z_K - E'}{2}$$

is an exceptional cycle. By Definition 5.3, $q \geq n + E'^2$ and $q \geq n + (E + E'')^2$, so

$$q \geq n + \frac{3E'^2 + (E + E'')^2}{4} = n + \frac{E'^2 + E''^2 + E^2}{2},$$

where the equality holds since $E = E' + E''$ implies that both fractions are equal to $E'^2 + E' \cdot E'' + E''^2$. Since $D^2 = Z_K \cdot D - 2\chi(D)$ for any exceptional cycle D and $E' + E'' + E = 2E$, we obtain

$$q \geq n + Z_K \cdot E - \chi(E') - \chi(E'') - \chi(E). \tag{5.1}$$

Since $E = E' + E''$, the definition of χ gives

$$\chi(E) = \chi(E') + \chi(E'') - E' \cdot E''.$$

Since E' and E'' have no common component on the smooth surface Y, $E' \cap E''$ is a finite set. Each point of $E' \cap E''$ has positive local intersection multiplicity, so $E' \cdot E'' \geq \#(E' \cap E'')$. By Lemma 5.1 for the reduced compact curves E' and E'', together with $\chi(\mathcal{O}_{E'}) = \chi(E')$ and $\chi(\mathcal{O}_{E''}) = \chi(E'')$,

$$2\chi(E') \leq \chi_{\mathrm{top}}(E') \quad \text{and} \quad 2\chi(E'') \leq \chi_{\mathrm{top}}(E'').$$

By Zariski's main theorem, E is connected. Since E has n irreducible components, $\chi_{\mathrm{top}}(E) = 1 - b_1(E) + n$. By inclusion–exclusion,

$$\chi_{\mathrm{top}}(E) = \chi_{\mathrm{top}}(E') + \chi_{\mathrm{top}}(E'') - \#(E' \cap E'').$$

Thus

$$\begin{aligned}\chi(E')+\chi(E'')+\chi(E) &= 2\chi(E')+2\chi(E'')-E'\cdot E''\\ &\le \chi_{\mathrm{top}}(E')+\chi_{\mathrm{top}}(E'')-\#(E'\cap E'')\\ &= 1+n-b_1(E).\end{aligned}$$

Together with (5.1), $q \ge n + Z_K\cdot E-(1+n-b_1(E))$, that is,

$$q \ge b_1(E)-1+Z_K\cdot E.$$

We prove (2). Since $\chi(\mathcal{O}_M)=\chi(M)$ and $h^0(\mathcal{O}_M)\ge 1$,

$$2h^1(\mathcal{O}_M)\ge 2+M^2-Z_K\cdot M.$$

Adding (1), we obtain

$$q+2h^1(\mathcal{O}_M)-b_1(E)\ge 1+M^2-Z_K\cdot(M-E).$$

Since $Z_K\cdot E_i\le 0$ for any i and $M\ge E$, $-Z_K\cdot(M-E)\ge 0$. By Lemma 5.2, $M^2\ge -\nu$. Thus

$$q+2h^1(\mathcal{O}_M)-b_1(E)\ge 1-\nu.$$

The proposition follows. □

6. Proof of the main theorem

Proof of Theorem 1.2. **Step 1.** In this step we reduce to singular germs with positive geometric genus and bound $\mu(f,0)-6p_g(X,0)$ from below for them.

If $(X,0)$ is smooth, then one of $\partial f/\partial x$, $\partial f/\partial y$, $\partial f/\partial z$ is a unit, thus $\mu(f,0)=0$. Since id_X is a resolution of X, $p_g(X,0)=0$. Hence

$$\mu(f,0)=6p_g(X,0)=0.$$

If $(X,0)$ is singular, then $f\in\mathfrak{n}^2$, so $\partial f/\partial x$, $\partial f/\partial y$, $\partial f/\partial z\in\mathfrak{n}$. Thus

$$\mu(f,0)=\dim_{\mathbb{C}} S/(\partial f/\partial x,\partial f/\partial y,\partial f/\partial z)\ge \dim_{\mathbb{C}} S/\mathfrak{n}=1.$$

In particular, $\mu(f,0)>6p_g(X,0)$ if $p_g(X,0)=0$.

Assume from now on that $(X,0)$ is singular and $p_g(X,0)>0$. Then f and $(X,0)$ satisfy Set-up 2.2, and we use the notation of Set-up 2.2, q as in Definition 5.3, and c, N_Q as in Definition 3.3. We write $\mu=\mu(f,0)$ and $p_g=p_g(X,0)$. Since Z_K is an exceptional cycle, by [KN17b, Lemma 7] there exists an exceptional cycle l such that

$$\frac{1}{2}Z_K\le l\le Z_K \quad\text{and}\quad \chi(l)=\min\{\chi(D): D \text{ is an exceptional cycle}\}.$$

Since $(Z_K-2D)^2=Z_K^2+4D\cdot(D-Z_K)=Z_K^2-8\chi(D)$ for any exceptional cycle D,

$$q=n+(Z_K-2l)^2=n+Z_K^2-8\chi(l). \tag{6.1}$$

Since $Z_K\ge l\ge Z_K-l\ge 0$, $I_K\subset I\cap J$ for $I=I(l)$ and $J=I(Z_K-l)$, so $IJ\supset I_K^2$ has finite colength. Let

$$\kappa=4(\lambda(I)+\lambda(J))-\lambda(IJ)-1.$$

For $g\in I$ and $g'\in J$, $\pi^*(gg')$ vanishes along $l+(Z_K-l)=Z_K$, thus $IJ\subset I_K$. Since $\lambda(I_K)=p_g$ by Lemma 2.4,

$$\lambda(IJ)\ge p_g.$$

By Lemma 2.3, (6.1), and Lemma 2.4 for $D=l$,

$$\begin{aligned}\mu-6p_g &= 6p_g+Z_K^2+n-b_1(E)=6p_g+q+8\chi(l)-b_1(E)\\ &\ge q-b_1(E)-2p_g+8(\lambda(I)+\lambda(J))\\ &= 2+q-b_1(E)+2\kappa+2(\lambda(IJ)-p_g),\end{aligned}$$

thus

$$\mu - 6p_g \geq 2 + q - b_1(E) + 2\kappa. \tag{6.2}$$

Step 2. In this step we prove $\mu - 6p_g > 0$ when $l = Z_K$.

Since $\pi_*\mathcal{O}_Y = \mathcal{O}_X$, $J = I(0) = R$. Thus $I = IJ = I_K$, $\lambda(I) = \lambda(IJ) = p_g$, $\lambda(J) = 0$, and $\kappa = 3p_g - 1$. By Proposition 5.4(2) and $h^1(\mathcal{O}_M) \leq p_g$ (Lemma 4.1),

$$2 + q - b_1(E) + 2\kappa = q - b_1(E) + 6p_g \geq 1 - \nu - 2h^1(\mathcal{O}_M) + 6p_g \geq 1 - \nu + 4p_g.$$

Since $f \in \mathfrak{n}^2$, $\nu \geq 2$. If $\nu = 2$, then $1 - \nu + 4p_g \geq 3$ as $p_g \geq 1$. If $\nu \geq 3$, then $p_g \geq \binom{\nu}{3}$ by Proposition 4.2, and

$$4\binom{\nu}{3} = \frac{2}{3}\nu(\nu-1)(\nu-2) \geq 2(\nu-1)$$

since $\nu(\nu-2) \geq 3$, thus $1 - \nu + 4p_g \geq \nu - 1 \geq 2$. In both cases (6.2) implies $\mu - 6p_g > 0$.

Step 3. In this step we assume $l \neq Z_K$ and bound the right side of (6.2) from below by N_I and $\lambda(I) - \lambda(J)$.

Since $l \geq Z_K - l \geq 0$ and $Z_K - l \neq 0$, $I \subset J \subset \mathfrak{m}$. Since $I_K \subset I$, I and J are $\mathfrak{m}$-primary. Let $\delta = \lambda(I) - \lambda(J) \geq 0$. By Proposition 3.4,

$$\kappa + 1 \geq c(IJ) + N_I + \lceil\sqrt{\delta}\rceil.$$

For any non-zero $t \in \mathbb{C}x + \mathbb{C}y + \mathbb{C}z$, $IJ + tR \subset I_K + tR$, thus $c(IJ) \geq c(I_K) = h^1(\mathcal{O}_M)$ by Lemma 4.1. Together with Proposition 5.4(2),

$$2 + q - b_1(E) + 2\kappa \geq q - b_1(E) + 2h^1(\mathcal{O}_M) + 2N_I + 2\lceil\sqrt{\delta}\rceil \geq 1 - \nu + 2N_I + 2\lceil\sqrt{\delta}\rceil. \tag{6.3}$$

For a non-zero ideal $Q \subset S$ let $\operatorname{mult}_0 Q = \min\{\operatorname{mult}_0 g \mid g \in Q\backslash\{0\}\}$. Since the product of the lowest degree forms of $g, g' \in S\backslash\{0\}$ is non-zero, $\operatorname{mult}_0(gg') = \operatorname{mult}_0 g + \operatorname{mult}_0 g'$. Let $\widetilde{I}, \widetilde{J} \subset S$ be the inverse images of I, J. The image of $\widetilde{I}\widetilde{J}$ in R is $IJ \subset I_K$, thus $\widetilde{I}\widetilde{J} \subset \mathfrak{n}^{\nu-2}$ by Proposition 4.2. Choosing $g \in \widetilde{I}$ and $g' \in \widetilde{J}$ with $\operatorname{mult}_0 g = \operatorname{mult}_0 \widetilde{I}$ and $\operatorname{mult}_0 g' = \operatorname{mult}_0 \widetilde{J}$, we have

$$\operatorname{mult}_0 \widetilde{I} + \operatorname{mult}_0 \widetilde{J} = \operatorname{mult}_0(gg') \geq \nu - 2.$$

Since $\widetilde{I} \subset \widetilde{J}$, $\operatorname{mult}_0 \widetilde{I} \geq \operatorname{mult}_0 \widetilde{J}$, so

$$2\operatorname{mult}_0 \widetilde{I} \geq \operatorname{mult}_0 \widetilde{I} + \operatorname{mult}_0 \widetilde{J} \geq \nu - 2.$$

Since $\mathfrak{m}^{N_I} \subset I$, $\mathfrak{n}^{N_I} \subset \widetilde{I}$, thus $\operatorname{mult}_0 \widetilde{I} \leq N_I$. Hence, letting $r = \lceil(\nu-2)/2\rceil$,

$$N_I \geq \operatorname{mult}_0 \widetilde{I} \geq r. \tag{6.4}$$

If $N_I = r$, then $\operatorname{mult}_0 \widetilde{I} = r$ by (6.4), so $\widetilde{I} \subset \mathfrak{n}^r \subset \widetilde{I}$ and $I = \mathfrak{m}^r$. If moreover $\delta = 0$, then $I = J$ as $I \subset J$. Thus

$$\text{if } N_I = r \text{ and } \delta = 0, \text{ then } I = J = \mathfrak{m}^r. \tag{6.5}$$

Let $u \in \mathfrak{m}$ be a general linear combination of the images of x, y, z. For any i, the linear combinations t of the images of x, y, z such that $v_i(t) > m_i$ form a proper linear subspace, since $\mathfrak{m}$ is generated by the images of x, y, z. Thus $v_i(u) = m_i$ for any i. If $N_I = r$ and $\delta = 0$, then $\mathfrak{m}^{2r} = IJ \subset I_K$ by (6.5), so $u^{2r} \in I_K$ and, by (4.1) for $D = Z_K$, $2rm_i = v_i(u^{2r}) \geq \operatorname{mult}_{E_i} Z_K$ for any i, that is,

$$Z_K \leq 2rM. \tag{6.6}$$

If $\nu = 2$, then (6.3) and $N_I \geq 1$ imply $2 + q - b_1(E) + 2\kappa \geq 1$. Thus we may assume $\nu \geq 3$, so $r \geq 1$ and $\nu \in \{2r+1, 2r+2\}$.

Step 4. In this step we prove $\mu - 6p_g > 0$ when $l \neq Z_K$ and $\nu = 2r + 1$.

By (6.3),

$$2 + q - b_1(E) + 2\kappa \geq 2(N_I - r) + 2\lceil\sqrt{\delta}\rceil.$$

By (6.4), $N_I - r$ and $\lceil\sqrt{\delta}\rceil$ are non-negative integers, so $2(N_I - r) + 2\lceil\sqrt{\delta}\rceil \geq 2$ if $N_I > r$ or $\delta > 0$. Thus we may assume $N_I = r$ and $\delta = 0$. By (6.5), $I = J = \mathfrak{m}^r$, so $\mathfrak{m}^{2r} = IJ \subset I_K$.

Since $v_i(u) = m_i$ for any i, $\operatorname{div}(\pi^* u) = M + H$ for some $H \geq 0$ having no exceptional component. Since $(M + H) \cdot E_i = \operatorname{div}(\pi^* u) \cdot E_i = 0$,

$$M \cdot E_i = -H \cdot E_i \leq 0 \text{ for any } i.$$

Let $W = 2rM - Z_K$, which is ≥ 0 by (6.6). Then $M \cdot W \leq 0$. Since $Z_K \cdot E_i \leq 0$ for any i and $M \geq E$, by Lemma 5.2,

$$-Z_K \cdot E \leq -Z_K \cdot M = -2rM^2 + M \cdot W \leq -2rM^2 \leq 2r\nu = 2r(2r+1).$$

Since $f \in \mathfrak{n}^{2r+1}$, $R/\mathfrak{m}^j \cong S/\mathfrak{n}^j$ for any $j \leq 2r+1$, so $\lambda(\mathfrak{m}^r) = \binom{r+2}{3}$ and $\lambda(\mathfrak{m}^{2r}) = \binom{2r+2}{3}$. Thus

$$\kappa = 8\binom{r+2}{3} - \binom{2r+2}{3} - 1 = \frac{2}{3}r(r+1)\big(2(r+2) - (2r+1)\big) - 1 = 2r(r+1) - 1.$$

By Proposition 5.4(1),

$$2 + q - b_1(E) + 2\kappa \geq 1 + Z_K \cdot E + 2\kappa \geq 1 - 2r(2r+1) + 4r(r+1) - 2 = 2r - 1 > 0.$$

By (6.2), $\mu - 6p_g > 0$.

Step 5. In this step we prove $\mu - 6p_g > 0$ when $l \neq Z_K$ and $\nu = 2r+2$.

By (6.3),

$$2 + q - b_1(E) + 2\kappa \geq -1 + 2(N_I - r) + 2\lceil\sqrt{\delta}\rceil,$$

and $-1 + 2(N_I - r) + 2\lceil\sqrt{\delta}\rceil \geq 1$ if $N_I > r$ or $\delta > 0$. Thus we may assume $N_I = r$ and $\delta = 0$. By (6.5), $I = J = \mathfrak{m}^r$, so $\mathfrak{m}^{2r} = IJ \subset I_K$. By Proposition 4.2, $I_K \subset \mathfrak{m}^{\nu-2} = \mathfrak{m}^{2r}$. Hence

$$I_K = \mathfrak{m}^{2r}.$$

In this case we bound $\mu - 6p_g$ by Lemma 2.3 directly. Since $f \in \mathfrak{n}^{2r+2}$, Lemma 2.4 implies

$$p_g = \lambda(\mathfrak{m}^{2r}) = \binom{2r+2}{3}.$$

For any non-zero $t \in \mathbb{C}x + \mathbb{C}y + \mathbb{C}z$, choosing linear coordinates of S with t as the first one,

$$R/(\mathfrak{m}^{2r} + tR) \cong S/(\mathfrak{n}^{2r} + tS) \cong \mathbb{C}\{y,z\}/(y,z)^{2r},$$

so $\lambda(\mathfrak{m}^{2r} + tR) = \binom{2r+1}{2}$. By Lemma 4.1,

$$h^1(\mathcal{O}_M) = c(I_K) = \binom{2r+1}{2}.$$

Let $W = 2rM - Z_K$, which is ≥ 0 by (6.6). Since $Z_K \cdot E_i \leq 0$ for any i and $W \geq 0$, $Z_K \cdot W \leq 0$. Since the intersection form on exceptional cycles is negative definite, $W^2 \leq 0$. Thus

$$4r^2M^2 = (Z_K + W)^2 = Z_K^2 + 2Z_K \cdot W + W^2 \leq Z_K^2,$$

and by Lemma 5.2,

$$Z_K^2 \geq 4r^2M^2 \geq -4r^2\nu = -4r^2(2r+2).$$

Since E is connected, $b_1(E) \leq 2h^1(\mathcal{O}_E)$ by the second inequality of Lemma 5.1. Since $M \geq E$, $H^1(\mathcal{O}_M) \to H^1(\mathcal{O}_E)$ is surjective. Thus

$$b_1(E) \leq 2h^1(\mathcal{O}_M) = 2\binom{2r+1}{2}.$$

By Lemma 2.3,

$$\begin{aligned}\mu - 6p_g &= 6p_g + Z_K^2 + n - b_1(E) \geq 6\binom{2r+2}{3} - 4r^2(2r+2) + n - 2\binom{2r+1}{2} \\ &= (8r^3 + 12r^2 + 4r) - (8r^3 + 8r^2) + n - (4r^2 + 2r) = n + 2r > 0.\end{aligned}$$

By **Steps 2–5**, $\mu > 6p_g$ if $(X, 0)$ is singular and $p_g > 0$. By **Step 1**, $\mu > 6p_g$ if $(X, 0)$ is singular and $p_g = 0$, and $\mu = 6p_g = 0$ if $(X, 0)$ is smooth. Thus $6p_g(X, 0) \leq \mu(f, 0)$, and equality holds if and only if $(X, 0)$ is smooth. □

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